Friction across contact materials

EMAE 285 · Lab 4

Experiment overview

Used an inclined plane to estimate static and kinetic friction for contact materials including felt and cork. Investigated onset of sliding and motion at different incline angles.

Variables and instrumentation

Incline angle, transit time, material, test repetition

Analysis and presentation

Friction coefficients by material and test condition

Engineering relevance

Useful for fixture design, material handling, conveyors and motion systems.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1BLiJtcMQxWko6HWWC_xOPolBbXBVvr0ooo2CCxCgqIc/edit